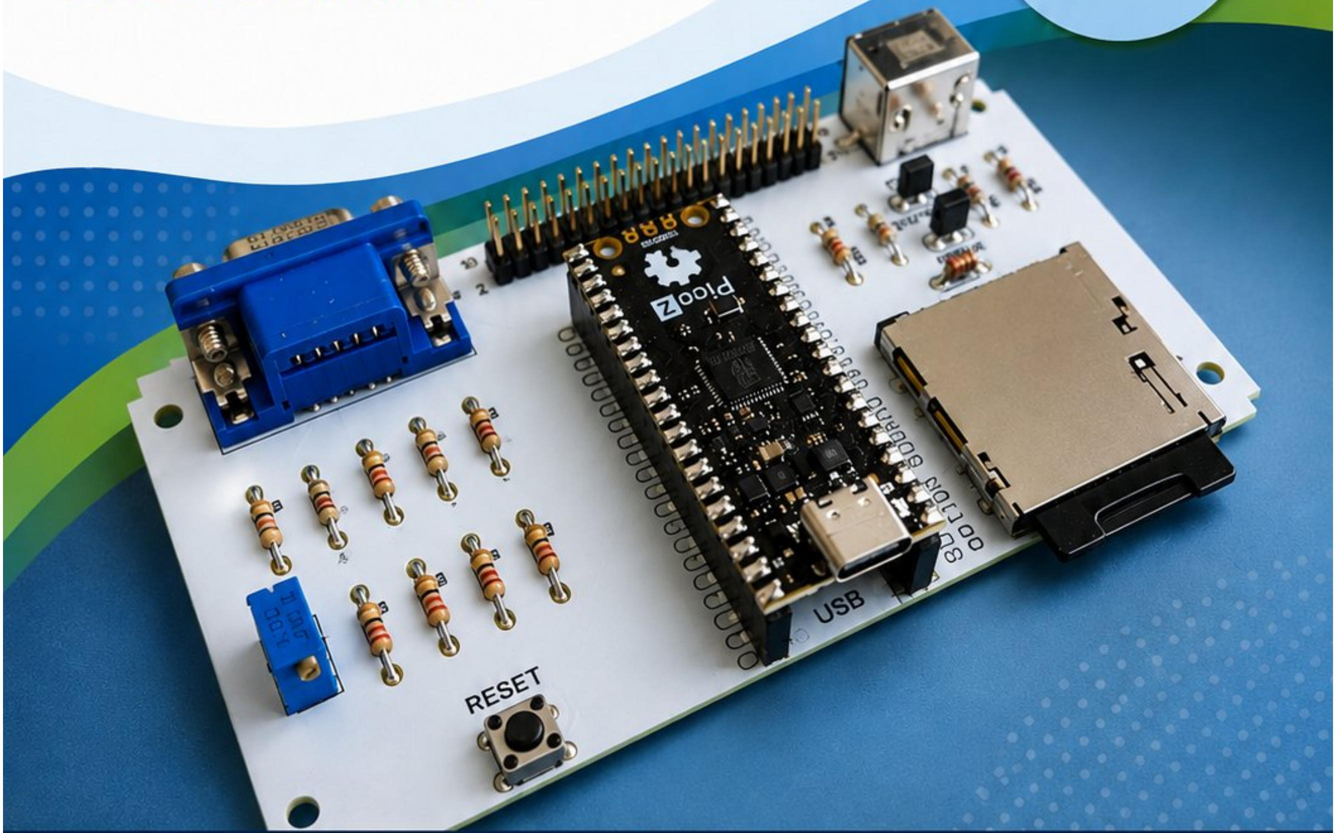
My First MMBasic Program

>_ Programming made easy!

A step-by-step introduction to MMBasic on the PicoMite and Colour Maximite

Perfect for beginners – no experience needed!



First steps with MMBasic

Enter & run your programs

Clear explanations and examples

Exercises & projects to try yourself

My first MMBasic Program

Step by Step into the World of MMBasic

for PicoMite and Colour Maximite

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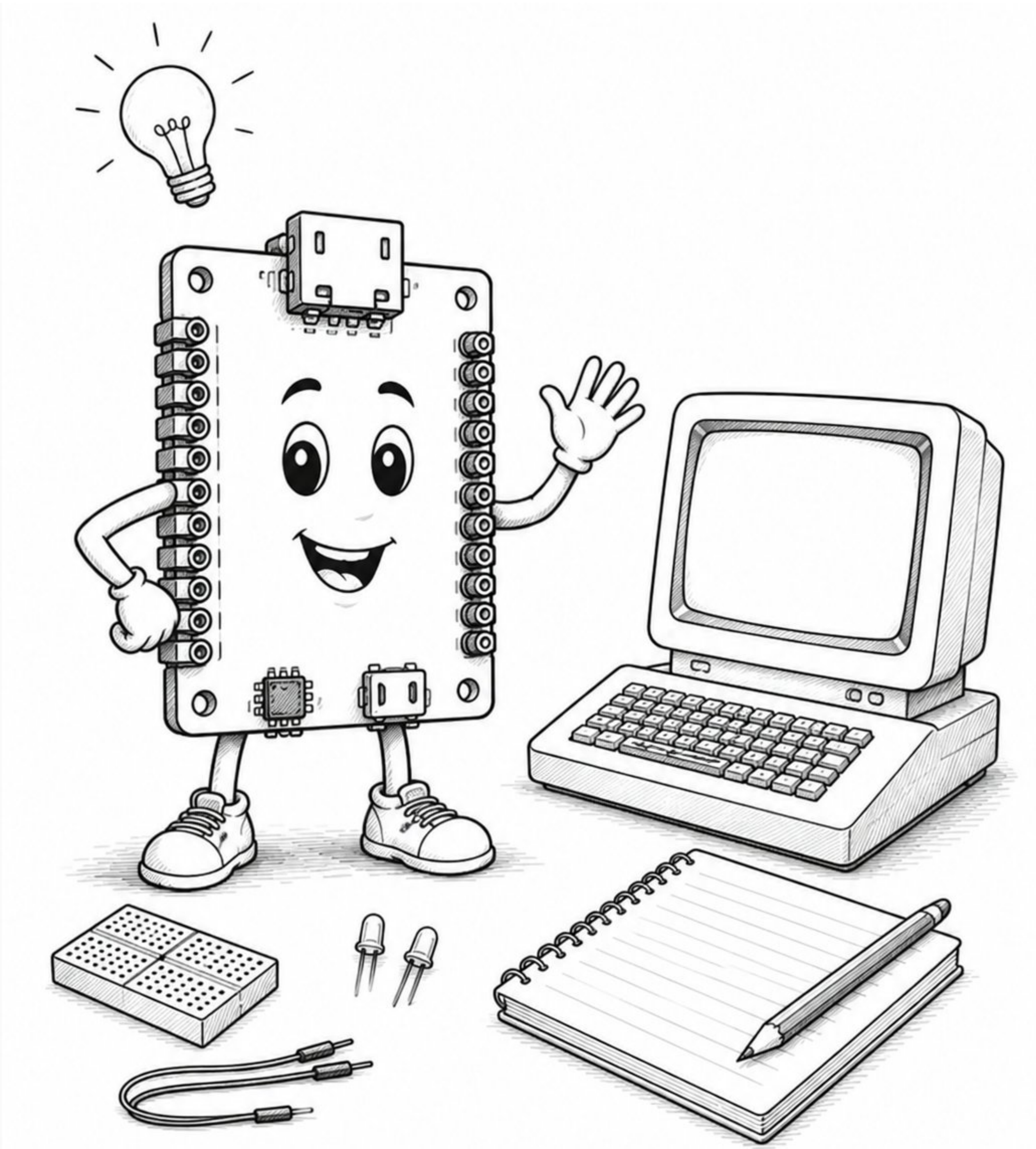


Legal Notice

Author: Manfred Becker

Project page:
https://github.com/ManiBecker/MyFirstMMBasicProgramm

This tutorial is being developed step by step and is publicly available on GitHub.

I would like to extend my special thanks to Geoff Graham, the developer of MMBasic, as well as Peter Mather, who played a pivotal role in further developing MMBasic for the Colour Maximite 2 and the PicoMite family.

Without their years of work and the freely available manuals, this tutorial would not be possible.

Note on the Creation of the Tutorial

This tutorial was created by the author and is continuously being updated.

Tools based on generative artificial intelligence (AI) were used to assist with drafting the text, revising the language, and structuring individual chapters.

All technical content, program examples, and technical statements were selected, reviewed, and approved for publication by the author.

Note on the Illustrations

Many of the illustrations in this tutorial were created with the assistance of generative artificial intelligence (AI).

Table of Contents

Welcome to Your MMBasic Journey

Welcome to **My First MMBasic Program**.

This tutorial is for anyone who wants to learn MMBasic and create their own programs.

It doesn't matter whether you already have programming experience or are using a programming language for the very first time. This book will guide you step by step through the fascinating world of MMBasic.

What is MMBasic?

MMBasic is a modern BASIC programming language designed for microcontrollers and computer systems.

It combines the simplicity of traditional BASIC dialects with the capabilities of today's hardware.

MMBasic is available for several platforms, including:

- PicoMite
- PicoMite VGA
- PicoMite HDMI
- Colour Maximite 2

With MMBasic you can develop much more than simple console applications. Typical projects include:

- Graphical applications
- Games
- Hardware projects
- Measurement and control systems
- Network applications

Whether you want to blink an LED, create your own game, or build a complete embedded project, MMBasic provides a powerful yet easy-to-learn environment.

The People Behind MMBasic

MMBasic was originally created by **Geoff Graham** for the Maximite, a small computer that boots directly into a BASIC command prompt.

Later, **Peter Mather** extended MMBasic to support several additional platforms, including the Colour Maximite 2 and the PicoMite family based on the Raspberry Pi Pico and Pico 2.

Without their many years of development, excellent documentation, and continuous firmware improvements, this tutorial would not have been possible.

Which MMBasic Platform Should You Choose?

One of MMBasic's greatest strengths is that it is not tied to a single computer.

Today it runs on a wide range of systems, from a simple Raspberry Pi Pico connected to a terminal all the way to powerful standalone computers with graphics, sound, USB, and SD card support.

If you're just getting started, the following overview may help you decide which platform best suits your needs.

Platform	Required Hardware	Graphics	GPIO	Best Suited For
MMBasic for Windows (MMB4W)	Windows PC	Yes	No	Learning the language without additional hardware.
MMBasic for Linux (MMB4L)	Linux PC or Raspberry Pi	Yes	No	Linux users who want to explore MMBasic.
DOS MMBasic	DOS PC or DOSBox	Text only	No	Classic DOS systems and DOSBox.
Raspberry Pi Pico + Terminal	Pico and USB cable	Text only	Yes	First experiments with hardware programming.

Platform	Required Hardware	Graphics	GPIO	Best Suited For
PicoMite VGA or HDMI	PicoMite system	Yes	Yes	Recommended platform for this tutorial.
Colour Maximite 2	Colour Maximite 2	Yes	Yes	A powerful all-in-one MMBasic computer.

Throughout this tutorial, the **PicoMite** will be used as the primary reference platform.

However, most example programs also run on other MMBasic systems with little or no modification.

> If your main goal is to learn the MMBasic language, you can begin immediately with **MMB4W**, **MMB4L**, or **DOS MMBasic** on your existing computer. The hardware projects in the later chapters require a system with GPIO pins, such as a PicoMite.

Choosing the Right Platform

If you're mainly interested in learning the MMBasic programming language, MMB4W, MMB4L, or DOS MMBasic are excellent starting points.

If you also want to connect LEDs, push buttons, sensors, and other electronic components, a **PicoMite** is the best choice. All hardware examples in this tutorial have been developed and tested on a PicoMite system.

If you already own a **Colour Maximite 2**, you'll also be able to run most examples without modification or with only minor adjustments.

Why This Tutorial?

Many programming books focus mainly on explaining individual commands and language features.

This tutorial takes a different approach.

Instead of learning isolated commands, we'll develop practical programs and projects together. Along the way, you'll discover the language naturally, one step at a time.

Starting with simple text output, we'll gradually work our way towards:

- Graphical applications
- Games
- Hardware projects
- Electronic experiments

The goal is not simply to memorize commands, but to understand how many small building blocks can be combined to create larger and more interesting programs.

Learning Through Practical Projects

Many of the examples in this tutorial are based on real-world projects and personal experience.

Among them are:

- Little Professor
- Pong
- Traffic Light Controller
- Hardware experiments using LEDs and push buttons
- Senso – The Electronic Memory Game

Each new project builds on concepts introduced in earlier chapters.

This gradual approach helps you develop a solid understanding of MMBasic while creating programs that become more interesting with every chapter.

How This Book Is Organized

This tutorial is designed to guide you step by step through the MMBasic programming language.

We'll begin with simple commands and small programs before moving on to more advanced topics such as:

- Variables
- Loops
- Subroutines and Functions
- Graphics Programming
- Games
- File Handling
- Hardware Projects
- Development Tools

Most chapters build upon concepts introduced earlier.

For that reason, beginners are encouraged to work through the tutorial in the order in which it is presented.

Working with MMBasic

One of MMBasic's most attractive features is its interactive command prompt.

Immediately after starting MMBasic, you'll see a prompt similar to this:

```
>
```

At this prompt, you can type BASIC commands and execute them immediately.

For example:

```
PRINT "Hello World"
```

After pressing **Enter**, MMBasic instantly displays the result:

```
Hello World
```

Many of the examples in the first chapters can therefore be tried directly at the command prompt without creating a complete program.

This immediate feedback is one of MMBasic's greatest strengths. You can experiment freely, try new commands, make mistakes, and instantly see what happens. It's one of the fastest and most enjoyable ways to learn a programming language.

Commands and Programs

In the first chapters, you'll often enter individual commands directly at the MMBasic prompt.

Later, you'll combine these commands into complete BASIC programs that can be saved, loaded, and improved over time.

This approach allows you to focus on one new concept at a time without becoming overwhelmed.

As your knowledge grows, so will your programs.

You'll soon discover that even complex applications are built from many small, easy-to-understand pieces.

Don't Be Afraid of Making Mistakes

One of the best ways to learn programming is through experimentation.

Try changing the examples.

Modify numbers.

Replace text.

Add your own ideas.

Sometimes your program won't work the way you expected—and that's perfectly normal.

Even experienced programmers make mistakes every day.

The important part isn't avoiding mistakes; it's understanding why they happened and learning how to fix them.

Every error teaches you something new.

Practice Is the Key

Reading about programming is useful.

Writing programs yourself is even better.

Whenever you finish a chapter, take a few minutes to experiment with the examples before moving on.

Ask yourself questions such as:

- What happens if I change this value?
- Can I make the program shorter?
- Can I add another feature?
- Can I improve the output?

Programming is a practical skill.

The more you experiment, the more confident you'll become.

Your Goal

By the time you've completed this tutorial, you won't just know the syntax of MMBasic—you'll understand how to solve problems with it.

You'll be able to:

- write your own programs
- create graphical applications
- develop simple games
- control electronic hardware
- structure larger projects
- continue learning on your own

Most importantly, you'll have the confidence to turn your own ideas into working programs.

Let's Get Started!

Programming is an exciting journey.

Every chapter introduces new concepts, new ideas, and new possibilities.

Take your time.

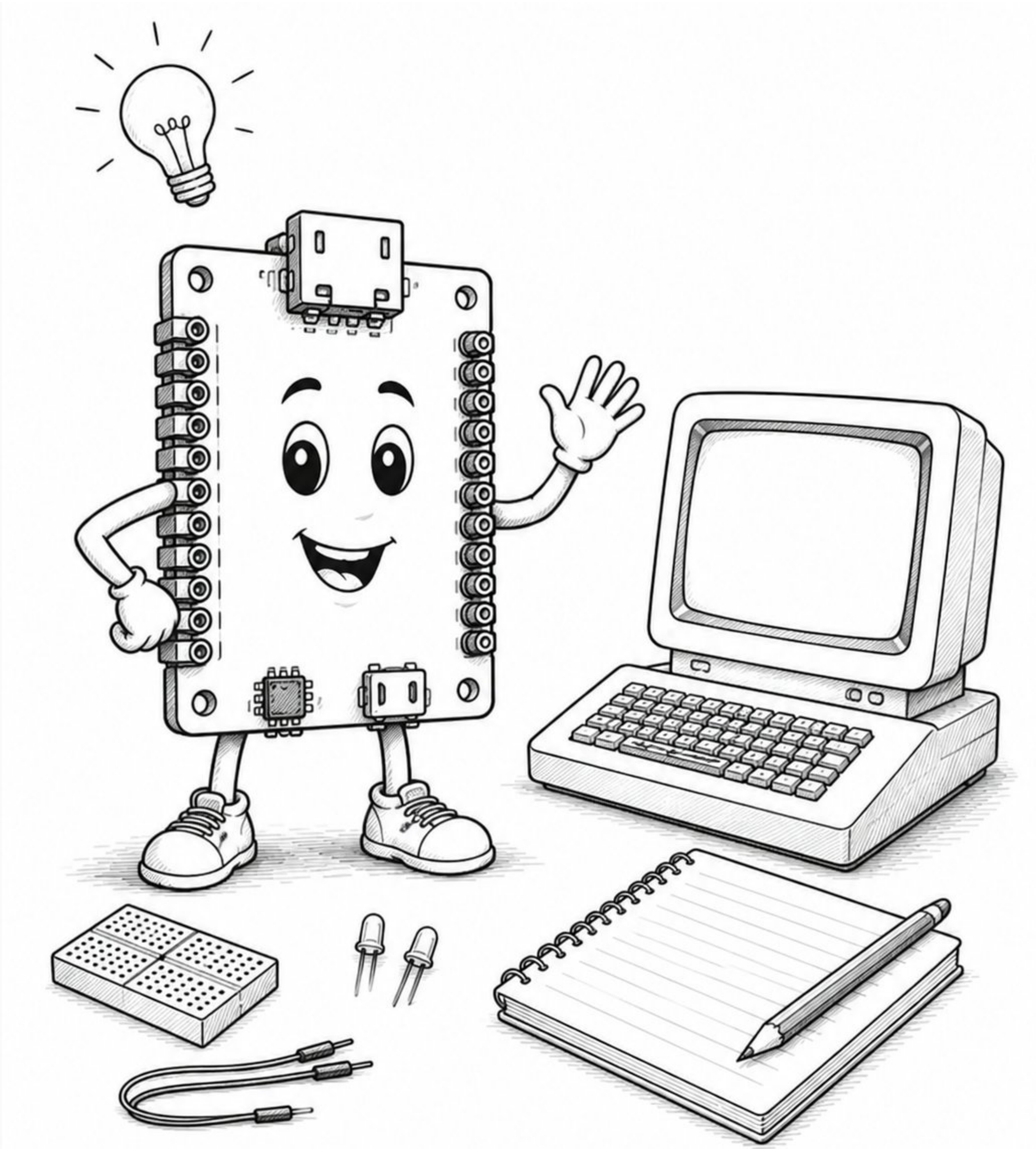
Experiment.

Have fun.

And remember:

Every experienced programmer once wrote their very first program.

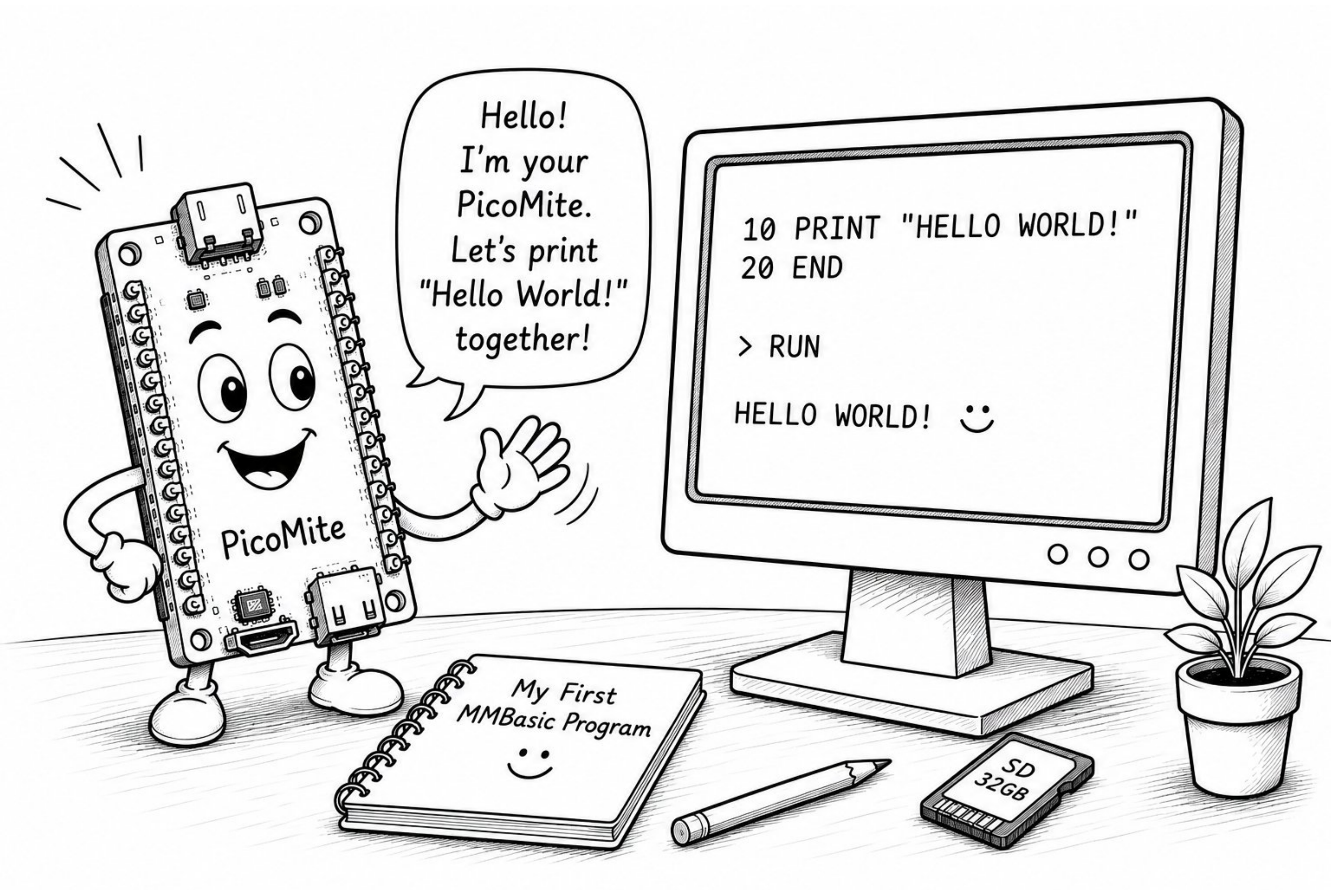
Let’s begin!



> "The best way to learn programming is to write programs."

Part I – Getting Started

Chapter 1. Hello World!



1.1. What You'll Learn

In this chapter, you'll learn:

- what a simple MMBasic program looks like
- how to run a program
- how to display text on the screen

By the end of this chapter, you'll have written and executed your very first MMBasic program.

1.2. Your First Program

Enter the following program:

```
PRINT "Hello World!"
```

Now run the program.

The screen will display:

```
Hello World!
```

Congratulations!

You've just written and executed your first MMBasic program.

1.3. Understanding the Program

Let's take a closer look:

```
PRINT "Hello World!"
```

The `PRINT` command displays text on the screen.

Everything enclosed in quotation marks is printed exactly as it appears.

You can replace the text with anything you like.

For example:

```
PRINT "Hello MMBasic!"
```

or

```
PRINT "I'm programming a PicoMite."
```

1.4. Experiment!

Now it's your turn.

Change the text between the quotation marks and see what happens.

Try examples such as:

```
PRINT "Hello Colour Maximite!"
```

```
PRINT "Today I'm learning MMBasic."
```

Can you think of your own message?

1.5. Try It Yourself

Complete the following exercises:

1. Display your name.
2. Display the name of your hometown.
3. Print a greeting for a friend.
4. Display three different messages one after another.

Example:

```
PRINT "Hello World!"
PRINT "Welcome to MMBasic!"
PRINT "Have fun programming!"
```

1.6. Summary

In this chapter, you've learned your first MMBasic command: `PRINT`.

The `PRINT` command displays text on the screen.

In the next chapter, you'll learn how to use MMBasic as a calculator and perform your first calculations.

1.7. Example Program

You can find the complete BASIC program here:

- 01_hello_world.bas

Part II – Building Larger Programs

Part III – Graphics and Games

Part IV – Hardware

Part V – Development Tools

Part VI – Files and Projects

Part VII – Advanced Topics

Epilogue

Appendix

MMBasic
TUTORIAL

YOUR FIRST STEP INTO MMBASIC – FROM YOUR FIRST PROGRAM TO YOUR OWN GAMES

This book guides you step by step through MMBasic, the programming language for the PicoMite and Colour Maximite 2.

From your first program to your own games – you'll learn everything you need to create your own projects.

EASY TO START
Clear explanations and many examples make getting started simple.

GRAPHICS, SOUND & SPRITES
Create impressive graphics, play sounds, and work with sprites.

GAMES & INTERACTIVE PROJECTS
Build games, animations and interactive programs – step by step.

DATA & PROJECTS
Save data, load projects and organize your programs.

YOUR OWN GAME IDEAS
Plan, develop and program your own games from the ground up.

> “Programming is more than code. It's the joy of **creating ideas** and **bringing them** to life.

Beginners or advanced users – this book shows you how to implement your ideas step by step and turn them into real MMBasic projects!

ABOUT THE AUTHOR

Manfred Becker has been programming for many years and has a passion for teaching and sharing knowledge. His goal is to make programming understandable and exciting – for beginners and experienced users alike.

github.com/ManfredBecker/MMBasic-Tutorial

Source code, examples and updates for this book